Topic 1 – Materials

1.1 Woods

Spec: Hardwoods – oak, mahogany, beech, jelutong, balsa; Softwoods – pine, cedar, larch, redwood

Key facts

• **Name the five hardwoods on the Edexcel spec.** Oak, mahogany, beech, jelutong, balsa.
• **Name the four softwoods on the Edexcel spec.** Pine, cedar, larch, redwood.
• **Why is jelutong used for models and pattern making?** Very fine, straight, even grain with few knots; soft and easy to carve; stable so it does not warp; takes paint well.
• **Why is beech used for toys, kitchen utensils and workbenches?** Hard, tough, close fine grain, no taste or smell, turns well and is very hard-wearing.
• **Which spec timber is naturally resistant to rot and insects, and used for cladding and shingles?** Western red cedar (also larch is durable outdoors).
• **What is the main disadvantage of oak contact with steel?** Tannic acid corrodes steel fixings and leaves black stains – use brass or stainless steel.
• **Why might mahogany not be considered sustainable?** Slow-growing tropical hardwood; logging (often illegal) causes deforestation and habitat loss; transported long distances, raising its carbon footprint.
• **Balsa: hardwood or softwood?** Hardwood (broad-leaved tree), though it is the lightest and softest commercial timber.
• **What moisture content is timber seasoned to for indoor furniture?** About 8–12% (kiln seasoning), roughly matching the indoor environment.
• **Give two properties of redwood that suit roof trusses.** Good strength-to-weight ratio, straight grained in long lengths, stable, relatively resistant to decay.

	Hardwoods	Softwoods
Tree	Broad-leaved, mostly deciduous (angiosperms)	Coniferous, mostly evergreen, needles and cones (gymnosperms)
Growth	Slow – often 80–150+ years	Fast – often around 30 years
Cost & supply	More expensive, limited sizes	Cheaper, long straight lengths
Typical uses	Quality furniture, flooring, tools, turning	Construction, joinery, cladding, DIY furniture

Timber	Appearance	Key working properties	Typical applications
Oak	Light brown, open grain, attractive figure (silver ray when quarter sawn)	Very strong, hard, tough, durable outdoors; contains **tannic acid** (corrodes steel fixings)	High-end furniture, flooring, beams, boat building, barrels
Mahogany	Reddish-brown, fine even grain, often interlocking	Fairly strong, stable, easy to work, turns and polishes beautifully; few knots	Quality furniture, veneers, turned legs, panelling, musical instruments
Beech	Pale pinkish-cream, close fine grain with flecks	Hard, tough, very close grain, steams and bends well, no taste or smell	Children's toys, kitchen utensils, chairs, workbenches, tool handles
Jelutong	Very pale cream, very fine even straight grain	Soft, easy to carve, stable, few knots, takes paint well	Pattern making (casting), architectural and prototype models, carving
Balsa	Very pale, open texture	Extremely light, very soft, low strength, easy to cut with a craft knife	Model aircraft, prototypes, rafts, insulation, model making

Timber	Appearance	Key working properties	Typical applications
Pine (Scots pine)	Pale cream/yellow, knots, resinous	Cheap, light, easy to work, moderately strong; knots can fall out	Construction framing, DIY furniture, joinery
Cedar (Western red cedar)	Reddish-brown, straight grain, aromatic	Lightweight, naturally resistant to rot and insects, weather resistant	Exterior cladding, sheds, roof shingles, garden furniture
Larch	Pale reddish-brown, distinct growth rings	Tough, durable, water resistant, harder than most softwoods	Boat planking, fencing, cladding, decking
Redwood (Scots pine from the Baltic)	Reddish-brown/cream	Good strength-to-weight ratio, straight grained, long lengths, stable	Roof trusses, joists, construction, joinery

Key terms

Hardwood	Timber from broad-leaved, usually deciduous trees with enclosed seeds (angiosperms). Generally slow growing, denser and more expensive. Balsa is a hardwood despite being very soft.
Softwood	Timber from coniferous, usually evergreen trees with needles and cones (gymnosperms). Faster growing (often 30 years), cheaper, more sustainable.
Grain	The direction and pattern of the wood fibres. Straight, fine, even grain machines and finishes well; interlocking grain resists splitting.
Seasoning	Drying timber to reduce its moisture content (about 12–15% for indoor use) so it is stable and less likely to warp, split or rot. Air seasoning or kiln seasoning.
Conversion	Sawing a log into usable boards – through-and-through (plain) sawing or quarter sawing.
Figure	The decorative pattern on the surface of timber (e.g. the silver ray figure of quarter-sawn oak).
Tannic acid	Acid in oak that corrodes and stains steel fixings – use brass or stainless steel screws.
FSC	Forest Stewardship Council – certifies timber from responsibly managed forests where trees are replanted.

Exam tip: "Balsa is a hardwood" and "larch is a deciduous softwood" are classic trick facts. The examiners also want a property **linked to the product**: "fine grain *so that* it gives a smooth finish on a turned leg".

1.2 Metals and alloys

Spec: Ferrous metals – mild steel, carbon steels, cast iron; Non-ferrous metals – aluminium, copper, zinc, tin; Alloys (ferrous and non-ferrous) – stainless steel, duralumin, brass

Key facts

• **What percentage of carbon is in mild steel?** About 0.15–0.35%.
• **What percentage of carbon is in cast iron?** About 2–4% (often 3.5%).
• **Why is cast iron used for lathe beds and engine blocks?** Very strong in compression, rigid, absorbs vibration, flows well into moulds, graphite makes it self-lubricating and easy to machine.
• **What is duralumin and where is it used?** Aluminium alloyed with ~4% copper, plus magnesium and manganese. High strength-to-weight ratio – aircraft frames, bike parts.
• **Name two metals alloyed with iron and carbon to make stainless steel.** Chromium and nickel (also manganese, molybdenum).
• **Why is brass used for 3-pin plug pins?** Good electrical conductor, harder and more wear resistant than copper, corrosion resistant, machines well.
• **Why is zinc used to coat steel (galvanising)?** Zinc is more reactive, so it corrodes sacrificially instead of the steel, even if the coating is scratched.
• **Main uses of tin?** Coating steel cans (tinplate) and as an alloying element in solder, bronze and pewter. Low melting point (232 °C).
• **Two key properties of copper?** Excellent electrical and thermal conductivity; ductile and malleable; corrosion resistant (forms green patina).
• **Why does aluminium resist corrosion?** It forms a thin protective oxide layer; this can be thickened by anodising.
• **Which carbon steel would you choose for a chisel blade and why?** High carbon (tool) steel – can be hardened and tempered so the edge stays sharp.

Metal	Composition	Properties	Uses
Mild steel (low carbon)	Iron + 0.15–0.35% C	Tough, ductile, malleable, good tensile strength, easily welded, cheap; **rusts**; cannot be hardened (only case hardened)	Car bodies, girders, nuts and bolts, gates, furniture frames
Medium carbon steel	0.35–0.7% C	Stronger and harder than mild steel, less ductile	Springs, garden tools, axles
High carbon steel / tool steel	0.7–1.5% C	Very hard, holds an edge, brittle unless tempered; can be hardened and tempered	Chisels, files, saw blades, drills, punches
Cast iron	2–4% C	Hard, brittle, strong in compression, flows well when molten, self-lubricating (graphite), absorbs vibration	Lathe and machine beds, vices, engine blocks, manhole covers, cookware

Metal	Properties	Uses
Aluminium	Light (density 2.7 g/cm³, ⅓ of steel), good strength-to-weight ratio, corrosion resistant (oxide layer), good conductor, easily cast and extruded, recyclable; low melting point ~660 °C	Drinks cans, foil, window frames, bike frames, aircraft, heat sinks, castings
Copper	Excellent electrical and thermal conductor, ductile, malleable, corrosion resistant, can be work hardened	Electrical wire, PCBs, pipes, heat exchangers, roofing
Zinc	Bluish-white, brittle at room temperature, very corrosion resistant, low melting point	Galvanising steel (sacrificial coating), die casting (Zamak), batteries, roofing
Tin	Soft, malleable, low melting point (232 °C), resists corrosion, non-toxic	Tinplate food cans (tin coating on steel), solder, alloying (bronze, pewter)

Alloy	Made of	Why it is better	Uses
Stainless steel (ferrous)	Iron + carbon + min. 11% chromium (+ nickel, manganese, molybdenum)	Chromium oxide film resists corrosion, hygienic, hard, attractive; harder to work and expensive	Sinks, cutlery, surgical tools, kitchen equipment
Duralumin (non-ferrous)	Aluminium + ~4% copper + magnesium + manganese	Much stronger and harder than pure aluminium while staying light; can be age hardened	Aircraft structures, vehicle panels, bike components
Brass (non-ferrous)	Copper ~65% + zinc ~35%	Harder than copper, gold coloured, machines and casts easily, corrosion resistant, good conductor	Plug pins, door furniture, musical instruments, valves, screws

Key terms

Ferrous metal	A metal containing iron. Most are magnetic and will rust unless protected (stainless steel is the main exception).
Non-ferrous metal	A metal that contains no iron, so it does not rust (it may still corrode/oxidise) and is usually non-magnetic.
Alloy	A mixture of a metal with one or more other elements (metals or carbon) to improve its properties, e.g. strength, hardness or corrosion resistance.
Mild steel	Low carbon steel, about 0.15–0.35% carbon. Tough, ductile, malleable, cheap, easily welded; rusts; cannot be hardened by heating and quenching.
Carbon steel	Medium (0.35–0.7%) and high (0.7–1.5%) carbon steels: harder and stronger but more brittle; can be hardened and tempered for tools.
Cast iron	Iron with 2–4% carbon. Hard, brittle, strong in compression, excellent fluidity for casting, self-lubricating graphite flakes.
Duralumin	Aluminium alloy with about 4% copper plus magnesium and manganese – much stronger than pure aluminium, used for aircraft structures.
Brass	Alloy of copper (about 65%) and zinc (about 35%). Gold colour, corrosion resistant, machines and casts well, good conductor.
Stainless steel	Steel alloyed with at least 10.5–11% chromium (often plus nickel). Chromium forms a self-repairing oxide layer that resists corrosion.
Work hardening	Metal becomes harder and more brittle when it is bent, hammered or rolled cold. Annealing restores ductility.

Exam tip (2024 Q1b): High carbon steel is chosen for a chisel because it is **hard** so the edge stays sharp – not because it is tough. Hardening makes it more brittle; tempering trades a little hardness for toughness.

1.3 Polymers

Spec: Thermoplastics – acrylic, polyethylene, PET, PVC, polypropylene (PP), ABS; Thermosetting plastics – epoxy resin (ER), urea formaldehyde (UF), polyester resin (PR); Elastomers – rubber

Key facts

- **Difference between thermoplastics and thermosets?** Thermoplastics soften on heating and can be remoulded/recycled; thermosets form permanent cross-links when cured and cannot be re-softened.
- **Properties and uses of acrylic (PMMA)?** Stiff, hard, transparent (92% light transmission), polishes well, weather resistant, but brittle and scratches. Signs, light diffusers, display cases, car lights.
- **Why is PET used for fizzy drinks bottles?** Transparent, tough, good tensile strength to resist pressure, impermeable to CO_2, food safe, blow moulds well, recyclable (code 1).
- **What is special about polypropylene?** Tough, good fatigue resistance so it forms integral hinges, chemical resistant, light; used for food containers, chair shells, ropes.
- **Properties and uses of ABS?** Tough, hard, scratch and impact resistant, high-gloss finish, takes colour; used for games controllers, phone cases, toys (LEGO), helmets, appliance casings.
- **HDPE vs LDPE?** HDPE is stiffer and harder (bottles, crates, pipes); LDPE is softer and more flexible (carrier bags, squeezy bottles, film).
- **Rigid vs flexible PVC?** Rigid uPVC: window frames, pipes, guttering. Flexible (plasticised) PVC: cable insulation, flooring, inflatables, clothing.
- **Uses of epoxy resin?** Strong adhesive (Araldite), coatings, encapsulating electronics, PCB boards and the matrix of carbon fibre composites.
- **Why is urea formaldehyde used for electrical sockets?** Thermoset so it will not soften if it gets hot, good electrical insulator, hard, rigid and can be white.
- **Polyester resin use?** Matrix for glass reinforced plastic (GRP) boat hulls and car panels; casting and embedding.
- **Why is natural rubber used for a sink plug?** Flexible elastomer that compresses to form a watertight seal; waterproof and resistant to detergents.

	Thermoplastics	Thermosetting plastics	Elastomers
Structure	Long chains, weak forces between them, no cross-links	Chains rigidly **cross-linked** into a 3D network during curing	Coiled chains with a few cross-links
On heating	Soften and can be reshaped repeatedly	Do not soften – char/burn at high temperature	Soften slightly; vulcanised rubber does not melt
Recycling	Can be remelted and recycled	Cannot be remelted (can be ground as filler)	Difficult (tyres reused as crumb)
Typical processes	Injection, blow and rotational moulding, extrusion, vacuum forming	Compression moulding, laminating (GRP/CFRP), casting	Moulding, extrusion, calendering

Polymer	Key properties	Applications
Acrylic (PMMA, "Perspex")	Stiff, hard, excellent clarity, polishes to a high gloss, weather/UV resistant, takes colour; **brittle**, scratches easily; laser cuts and line bends well	Signs, displays, light diffusers, car rear lights, baths, school projects
Polyethylene (polythene) – HDPE / LDPE	HDPE: stiff, hard, tough, chemical resistant. LDPE: soft, flexible, tough. Both cheap, waterproof, food safe	HDPE: bottles, crates, pipes, bins. LDPE: carrier bags, squeezy bottles, film, cable insulation
PET (polyethylene terephthalate)	Transparent, tough, strong, impermeable to gas, food safe, widely recycled (code 1)	Drinks bottles, food packaging, polyester fibre (fleece)
PVC (polyvinyl chloride)	Rigid uPVC: stiff, hard, weather and chemical resistant. Flexible PVC (with plasticiser): soft, waterproof. Cheap; burning releases toxic gases	uPVC window frames, pipes, guttering; flexible PVC cable insulation, flooring, inflatables, toys
Polypropylene (PP)	Tough, light, excellent fatigue resistance (integral hinges), good chemical and heat resistance (to ~100 °C), food safe	Food containers with hinged lids, chair shells, ropes, kettles, medical equipment, chisel handles
ABS	Tough, hard, scratch and impact resistant, high-quality glossy finish, colourful	Games controllers, phone cases, kitchen appliances, toys, helmets, car trims

Polymer	Key properties	Applications
Epoxy resin (ER)	Strong, rigid, excellent adhesion, chemical and heat resistant; two-part (resin + hardener)	Adhesive (Araldite), coatings, encapsulating electronics, PCBs, matrix for CFRP
Urea formaldehyde (UF)	Hard, rigid, brittle, heat resistant, excellent electrical insulator, light colours	Electrical sockets and plugs, switches, wood adhesives (MDF, chipboard)
Polyester resin (PR)	Stiff, hard, brittle alone; cheap; cures with a catalyst; good chemical resistance	Matrix for GRP (boat hulls, car panels), casting, embedding objects

Key terms

Polymer	A material made of long chain molecules (made from repeating units called monomers). Most are made from crude oil.
Thermoplastic	A polymer with long chains and weak bonds between them; it softens when heated and can be reshaped and recycled many times.
Thermosetting plastic	A polymer whose chains become permanently cross-linked when it sets (cures). It cannot be re-softened by heat.
Elastomer	A polymer with coiled, lightly cross-linked chains that can be stretched a lot and return to shape, e.g. natural rubber, neoprene, silicone.
Vulcanisation	Heating rubber with sulfur to form cross-links, making it harder, tougher and more elastic.
Integral (living) hinge	A thin flexible section moulded into polypropylene that can be flexed thousands of times without breaking.
Resin identification code	Numbered triangle on plastic products (1 PET, 2 HDPE, 3 PVC, 4 LDPE, 5 PP, 6 PS, 7 other) to help sorting for recycling.
Plasticiser	Additive that makes a polymer such as PVC softer and more flexible.

> **Exam tip:** When explaining why a polymer suits a product, use the chain: **property → what it does in use → benefit to the user**. "PP is tough *so* the chisel handle will not split when hit with a mallet, *which* makes it safer and last longer."

1.4 Composites

Spec: Carbon fibre (CFRP), glass fibre (GRP); Medium Density Fibre Board (MDF), hardboard, chipboard, plywood

Key facts

- **What is a composite material?** Two or more materials combined (reinforcement in a matrix) to give improved properties.
- **Key properties of CFRP?** Extremely high strength-to-weight ratio, stiff, tough, corrosion resistant, can be anisotropic; expensive and hard to recycle.
- **Key properties of GRP?** Good strength-to-weight ratio, tough, waterproof, does not rot or corrode, moulds into large complex one-piece shapes; cheaper than CFRP.
- **Why is GRP good for a boat hull?** Moulded as one seamless piece so it is watertight; light but strong; does not rot; low maintenance; can be repaired with patch kits.
- **How is plywood made and why is it strong?** Odd number of veneers glued with grain at 90° to each layer – strong in all directions and resists warping/splitting.
- **What is MDF made from?** Fine wood fibres bonded with urea formaldehyde resin under heat and pressure.
- **Advantages of MDF?** Smooth, flat, uniform, no grain or knots, machines and paints well, stable, cheap, large sheets.
- **Disadvantages of chipboard?** Weak, swells and crumbles if wet, edges chip, screws hold poorly, heavy.
- **What is hardboard used for?** Cheap backs of furniture/drawer bottoms, door skins – thin, smooth on one side, textured on the other.
- **Why is MDF dust a hazard?** Very fine dust and formaldehyde resin – use extraction and a dust mask.

	Carbon fibre (CFRP)	Glass fibre (GRP)
Made from	Carbon fibre cloth/tow in epoxy resin	Glass fibre mat or woven cloth in polyester (or epoxy) resin
Properties	Very high strength-to-weight ratio, very stiff, tough, fatigue and corrosion resistant, distinctive woven look, can be engineered **anisotropic**	Good strength-to-weight ratio, tough, waterproof, does not rot or corrode, good electrical insulator
Drawbacks	Very expensive, labour intensive, difficult to recycle, can fail suddenly	Itchy fibres and styrene fumes (COSHH), surface can crack/craze, hard to recycle
Uses	F1 cars, racing bikes, helmets, aircraft parts, tennis racquets, prosthetics	Boat hulls, canoes, car body panels, water tanks, wind turbine blades

Board	Made from	Properties	Uses
MDF	Fine wood fibres + urea formaldehyde resin, pressed under heat	Smooth, flat, uniform, no grain, stable, machines and paints well, cheap; swells if wet (moisture-resistant grades exist), heavy, dust is hazardous	Flat-pack furniture, kitchen units, model bases, shop fittings
Hardboard	Compressed wood fibres (wet process), thin	Cheap, smooth on one face, textured on the other, not very strong, swells if wet	Furniture backs, drawer bottoms, clipboard backs, door skins
Chipboard	Wood chips/particles bonded with resin	Cheap, flat, stable, uses waste wood; weak, edges crumble, swells in water, screws hold poorly – special fittings needed	Kitchen worktops and carcasses (melamine or veneer faced), flooring, flat-pack
Plywood	Odd number of veneers glued with grain at 90°	Strong in all directions, resists splitting and warping, can be bent (flexiply); exterior and marine grades	Furniture, boat building, shelving, construction, skateboards

Key terms

Composite	A material made from two or more different materials bonded together, giving better combined properties than either alone.
Matrix	The binder (e.g. resin, glue) that surrounds and supports the reinforcement and transfers load to it.

Reinforcement	The strong, stiff part of a composite (fibres, particles or sheets) that carries the load.
Anisotropic	Having different properties in different directions. Fibre composites can be laid up so they are strongest where the load is.
Lay-up	Building a fibre composite by placing layers of mat or woven cloth into a mould and wetting with resin.
Manufactured board	Man-made sheet material made from timber veneers, particles or fibres bonded with adhesive – e.g. MDF, plywood, chipboard, hardboard.
Veneer	A thin slice of timber (0.5–3 mm). Used for plywood layers or as a decorative facing on cheaper boards.
Lipping	A strip of solid wood or edging tape glued to the edge of a board to hide and protect the core.

Exam tip (2023 Q1): For veneered chipboard, link each property to the product: large flat sheets *so* no joints are needed in a wardrobe door; cheaper than solid timber *so* the flat-pack can be sold at a budget price; but the core swells if damp and screws pull out on re-assembly.

1.5 Papers and boards

Spec: Drawing papers – layout, tracing, copier, cartridge; Commercial printing papers – bond, coated; Boards – mounting board, corrugated board, foam board, folding box board, foil-lined board

Key facts

• **Layout paper – properties and use?** Thin (~50 gsm), smooth, semi-translucent, cheap; quick sketching and developing ideas by tracing over; markers bleed through.
• **Tracing paper – properties and use?** Translucent, smooth, fairly strong (60–90 gsm); copying and transferring drawings and overlays.
• **Copier paper – properties and use?** ~80 gsm, smooth, opaque, bright white, cheap; printing, photocopying, everyday drawing.
• **Cartridge paper – properties and use?** Heavy (120–150 gsm+), slightly textured, opaque, tough; sketching, pencil, ink, watercolour and presentation drawings.
• **What is bond paper used for?** Good-quality, strong uncoated paper (~80–120 gsm) for letterheads, forms and documents – originally for bonds/certificates.
• **Why is coated paper used for magazines and brochures?** The smooth clay coating stops ink soaking in, giving sharp, bright, high-quality images (gloss, silk or matt finishes).
• **Corrugated board – structure and use?** Fluted paper sandwiched between liners; strong, rigid, cushioning, light, cheap, recyclable – transit/shipping boxes, packing cases.
• **Foam board – structure and use?** Polystyrene foam core between two paper faces; very light, rigid, easy to cut – architectural models, display/presentation boards.
• **Folding box board (FBB) – properties?** Multi-layer board with bleached outer layers: stiff, light, excellent printing surface, scores and folds without cracking – cereal and cosmetic boxes.
• **Foil-lined board – why for a drinks carton?** Aluminium foil layer is a barrier to moisture, air and light; inert so it does not taint the drink; prints well on the outside.
• **Mounting board – use?** Thick, rigid, often coloured board used to mount and frame artwork/photographs and for presentation.

Paper	Typical weight	Properties	Uses
Layout	~45–50 gsm	Thin, smooth, semi-translucent, cheap; marker ink can bleed through	Initial sketching and developing ideas by tracing earlier drafts
Tracing	60–90 gsm	Translucent (treated with acid/oil or heavily refined), smooth, strong for its weight, takes pencil and ink	Copying/transferring drawings, overlays, technical drawings
Copier	~80 gsm	Opaque, smooth, bright white, cheap, consistent	Printing, photocopying, everyday drawing and notes
Cartridge	120–220 gsm	Heavy, opaque, slightly textured ("tooth") surface, tough; takes pencil, ink, pastel and light watercolour without cockling	Sketching, rendering, presentation drawings

Paper	Properties	Uses
Bond	Strong, durable, uncoated, high quality, takes ink well, often watermarked (80–120 gsm)	Letterheads, official documents, forms, envelopes
Coated (gloss, silk, matt)	Surface coating of china clay/calcium carbonate fills the gaps between fibres, so ink sits on top – sharp, bright, high-resolution images; less absorbent	Magazines, brochures, catalogues, photographic prints, leaflets

Board	Structure	Properties	Uses
Mounting board	Thick, rigid, dense card, often coloured, with a bevel-cut window	Rigid, stable, clean surface, can be acid-free to protect artwork	Mounting and framing artwork and photographs, presentation
Corrugated board	Fluted paper glued between flat **liners** (single or double wall)	Strong, rigid, impact absorbing (flutes act like arches), light, cheap, recyclable and often recycled	Transit/shipping boxes, packing cases, pizza boxes, display stands

Board	Structure	Properties	Uses
Foam board	Polystyrene (foam) core between two paper or card faces	Very light, rigid, smooth printable surface, easy to cut with a knife; dents easily, not easily recycled	Architectural and prototype models, display and presentation boards, signs
Folding box board (FBB)	Multi-layer: mechanical pulp core between bleached chemical pulp layers	Stiff for its weight, excellent smooth printing surface, scores/creases and folds without cracking, food safe	Cereal, cosmetic, medicine and food boxes, cartons
Foil-lined board	Board laminated with a thin aluminium foil layer (often plus polyethylene)	Barrier to moisture, air, light and grease; reflects heat; inert; prints well outside	Drinks cartons (Tetra Pak), takeaway containers, ready-meal lids, premium packaging

Key terms

gsm	Grams per square metre – the weight of paper or board. Paper is usually below about 200 gsm; board is heavier/thicker.
Microns	Thickness of board is often measured in microns (µm); 1000 µm = 1 mm.
Bleed	When ink from a marker or pen spreads into or through the paper.
Coated paper	Paper with a surface coating (e.g. china clay/calcium carbonate) giving a smooth gloss, silk or matt finish that holds sharp, bright printed images.
Flute	The wavy middle layer of corrugated board that gives rigidity and cushioning.
Virgin fibre	New wood pulp fibres (not recycled) – stronger, longer fibres, brighter.
Barrier layer	A layer such as aluminium foil or polymer film that stops moisture, air, light or grease passing through.

Exam tip (2020 Q2a, 2023 Q2a): You will not get credit for repeating a property in the stem ("strength" was given). Recall precise names – *folding box board, corrugated board* – rather than "card".

1.6 Textiles

Spec: Natural fibres – cotton, linen, wool; Manmade fibres – nylon, polypropylene, polyester; Textile treatments – flame resistant, polytetrafluoroethylene (PTFE)

Key facts

• **Name the three natural fibres on the spec and their sources.** Cotton (cotton plant seed pod), linen (flax plant stem), wool (sheep's fleece).
• **Name the three manmade fibres on the spec.** Nylon (polyamide), polypropylene, polyester (PET).
• **Key properties of cotton?** Soft, absorbent, comfortable, breathable, strong (stronger wet), washes well; creases, shrinks,

slow to dry, burns easily.
• **Key properties of linen?** Very strong, cool, absorbent, dries quickly, lustrous; creases badly, stiff; good for hot climates.
• **Key properties of wool?** Warm (traps air), absorbent but feels dry, elastic and crease resistant, naturally flame resistant; can felt and shrink, may itch.
• **Key properties of nylon?** Very strong, elastic, abrasion resistant, lightweight, quick drying, low absorbency; melts with heat, static.
• **Key properties of polyester?** Strong, crease resistant, low absorbency so dries quickly, resists stretching/shrinking, UV resistant, can be recycled PET.
• **Why is polypropylene used for ropes and thermal base layers?** Very light (floats), does not absorb water, wicks sweat, strong, resists chemicals and rot.
• **How does Gore-Tex (expanded PTFE) work?** Membrane has billions of pores per cm^2 – smaller than water droplets but larger than water vapour molecules, so it is waterproof yet breathable.
• **Where are flame resistant textiles needed?** Upholstered furniture (UK fire regulations), children's nightwear, workwear for welders/firefighters, curtains in public buildings.

Fibre	Source	Properties	Uses
Cotton	Seed pod (boll) of the cotton plant	Soft, absorbent, breathable, cool, strong (stronger wet), hard-wearing, washes well; creases, shrinks, slow drying, flammable. Needs lots of water and pesticides to grow (organic cotton is better)	T-shirts, jeans, towels, bedding, cushion covers
Linen	Stem of the flax plant	Very strong, cool and absorbent, dries fast, lustrous, gets softer with washing; creases easily, quite stiff	Summer clothing, tablecloths, tea towels, upholstery
Wool	Fleece of sheep (and goats, alpacas)	Warm (crimped fibres trap air), absorbs moisture without feeling wet, elastic, crease resistant, naturally flame resistant; can shrink and felt, may itch, needs careful washing	Jumpers, suits, carpets, blankets, insulation
Nylon (polyamide)		Very strong, elastic, abrasion resistant, lightweight, low absorbency so quick drying; melts at high heat, builds up static	Ropes, cords, tents, tights, carpets, sportswear, bags
Polyester		Strong, durable, crease resistant, does not stretch or shrink, low absorbency, quick drying, UV and mildew resistant; can be made from recycled PET bottles	Garden canopies, sportswear, fleeces, wadding, sails, blended with cotton
Polypropylene		Very light (floats), does not absorb water, wicks sweat, resists chemicals, rot and stains	Ropes, thermal base layers, carpets, non-woven bags, geotextiles

Key terms

Fibre	A fine hair-like strand, the raw material of textiles. Staple fibres are short; filaments are continuous.
Yarn	Fibres twisted (spun) together into a continuous thread for weaving or knitting.
Woven fabric	Made by interlacing warp and weft yarns at right angles – stable, strong, frays at cut edges.
Knitted fabric	Made by interlocking loops of yarn – stretchy, comfortable, may ladder.
Non-woven fabric	Fibres bonded or felted together by heat, adhesive or needling – e.g. felt, interfacing.
Breathable	Allows water vapour (sweat) to escape while keeping liquid water out.
PTFE	Polytetrafluoroethylene – a fluoropolymer. As a stretched (expanded) microporous membrane it makes fabric waterproof, windproof and breathable (Gore-Tex). As a coating it repels water, oil and stains.
Flame resistant treatment	Chemical finish (e.g. Proban) or inherently flame resistant fibres that stop a fabric igniting easily or make it self-extinguishing.

Exam tip (2021 Q8): When evaluating a PTFE jacket, link the material to the user: breathable *so* sweat escapes during exertion *keeping the walker dry and comfortable*; waterproof and windproof *so* they stay protected in inclement weather. Then weigh up cost, reproofing, and environmental concerns about fluoropolymers.

1.7 Smart and modern materials

Spec: Thermo-ceramics; Shape memory alloys (SMA); Reactive glass; Liquid crystal displays (LCD); Photo-chromic materials; Thermo-chromic materials; Quantum tunnelling composites (QTC)

Key facts

• **Define a smart material.** A material whose properties change reversibly in response to a stimulus such as temperature, light, stress or electric current.
• **What is a thermo-ceramic and give two uses?** Ceramic + metal powders sintered together; stays hard and stable at very high temperatures (~1000 °C) – turbine blades, cutting tool tips, engine parts, high-performance bearings.
• **Two drawbacks of thermo-ceramics?** Brittle – can shatter if dropped; expensive to develop and produce.
• **How does an SMA spectacle frame benefit the user?** Superelastic – bends and springs back, survives being sat on; deformed frames return to shape when heated; light and comfortable.
• **Give three uses of SMA.** Spectacle frames, medical stents, orthodontic braces, muscle wire actuators, thermostats, fire sprinkler valves.
• **What is reactive (electrochromic) glass?** Glass that switches between clear and opaque/tinted instantly when an electric current is applied – privacy windows, auto-dimming mirrors, welding visors.
• **Drawbacks of reactive glass?** Expensive to buy and install, needs constant power to stay in one state, reverts if power fails, limited service life (10–20 years).
• **How does an LCD work?** Liquid crystals between two polarising glass layers twist or untwist when a voltage is applied, controlling how much backlight passes through each pixel.
• **Benefits of LCD televisions?** Thin and light, energy efficient, low heat output, high resolution, no screen burn-in.
• **Drawbacks of LCDs?** Limited viewing angle, motion blur, backlight bleed/cannot show true black, fragile, lose contrast in heat.
• **Photo-chromic material – stimulus and uses?** Changes colour/darkens in UV light: transition lenses, novelty T-shirts, UV indicator labels, nail varnish.
• **Thermo-chromic material – stimulus and uses?** Changes colour at a set temperature: baby spoons, kettles, mugs, bath toys, forehead thermometers, battery testers.
• **How does QTC work?** Metal particles in an elastomer binder: no pressure = insulator; squeezed = particles close enough for electrons to 'tunnel', resistance falls dramatically (up to 10^{12}).
• **Three uses of QTC.** Variable-speed power tool triggers, pressure-sensitive touch pads/game controllers, robot grippers, prosthetic fingertips, wearable textile switches.

Material	Stimulus → response
Thermo-chromic	Temperature → colour
Photo-chromic	UV light → colour/darkness
Shape memory alloy	Temperature (or stress) → shape
Reactive (electrochromic) glass	Electric current → opacity/tint
QTC	Pressure → electrical resistance
LCD	Voltage → alignment of liquid crystals (light passes or is blocked)

Key terms

Smart material	A material with a property that changes reversibly in response to an external stimulus (heat, light, stress, electricity, pH) and changes back when the stimulus is removed.
Modern material	A material developed through new technology for improved performance (e.g. LCDs, thermo-ceramics), not necessarily reacting to a stimulus.

Nitinol	The most common shape memory alloy – nickel + titanium.
Transition temperature	The temperature at which an SMA changes phase and returns to its 'remembered' shape, or a thermo-chromic pigment changes colour.
Superelasticity	Ability of some SMAs to undergo very large elastic strain and spring back at room temperature (e.g. Memoflex glasses frames).
Sintering	Compacting powder in a mould and heating it to just below its melting point so particles fuse – used to make thermo-ceramics.
Electrochromic glass	Reactive glass that changes from clear to opaque/tinted when a voltage is applied (or removed).
Silver halide	Crystals in photo-chromic glass that darken in UV light.

Exam tip: Your teachers' slides also use "reactive glass" for photo-chromic lenses (silver halide crystals that darken in UV). In the exam, if the question separates them (2024 Q10), reactive glass = **electrically switched**.